

NANYANG JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

BIOLOGY

9744/04

Paper 4 Practical

27 August 2018

Candidates answer on the Question Paper

Additional Materials: As listed in the Confidential Instructions

2 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

Shift
Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	21
2	21
3	13
Total	55

This document consists of **22 printed pages and no blank page**.

[Turn over

Answer **all** the questions.

- 1** You are required to estimate the water potential of potato tissue.

When a piece of potato is put into a sucrose solution, as shown in Fig. 1.1, water will move by osmosis into and out of the potato cells. The net direction of movement of water depends on the difference in water potential between the potato cells and the sucrose solution.

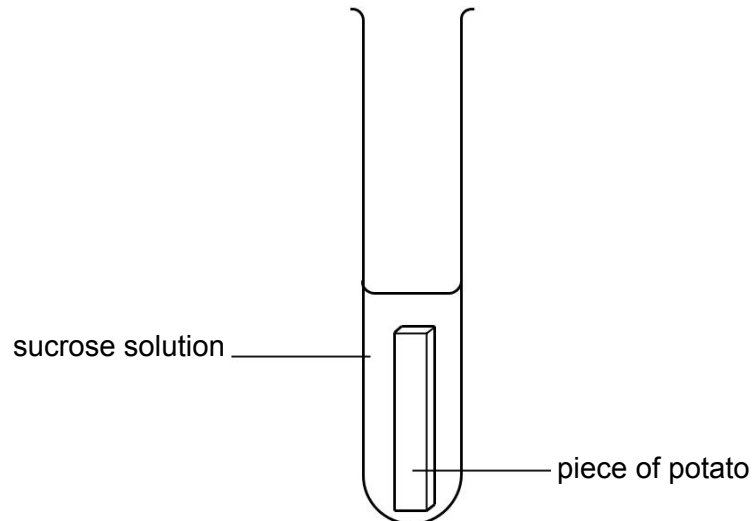


Fig. 1.1

- (a)** (i) Complete the sentences by using **two** of the following words.

gain

less

lose

more

If the potato cells water then the sucrose solution will become more dilute.

This will change the sucrose solution so that it becomes dense.

[1]

lose + less ;

A piece of potato is left in a sucrose solution for 15 minutes, to allow time for osmosis to take place.

The concentration of the sucrose solution after 15 minutes may be different from the original concentration.

After 15 minutes a blue dye is added to the sucrose solution to make a blue solution. The blue dye does not affect the concentration of the sucrose solution.

A small volume of this blue solution is then released into a known concentration of sucrose solution as shown in Fig. 1.2.

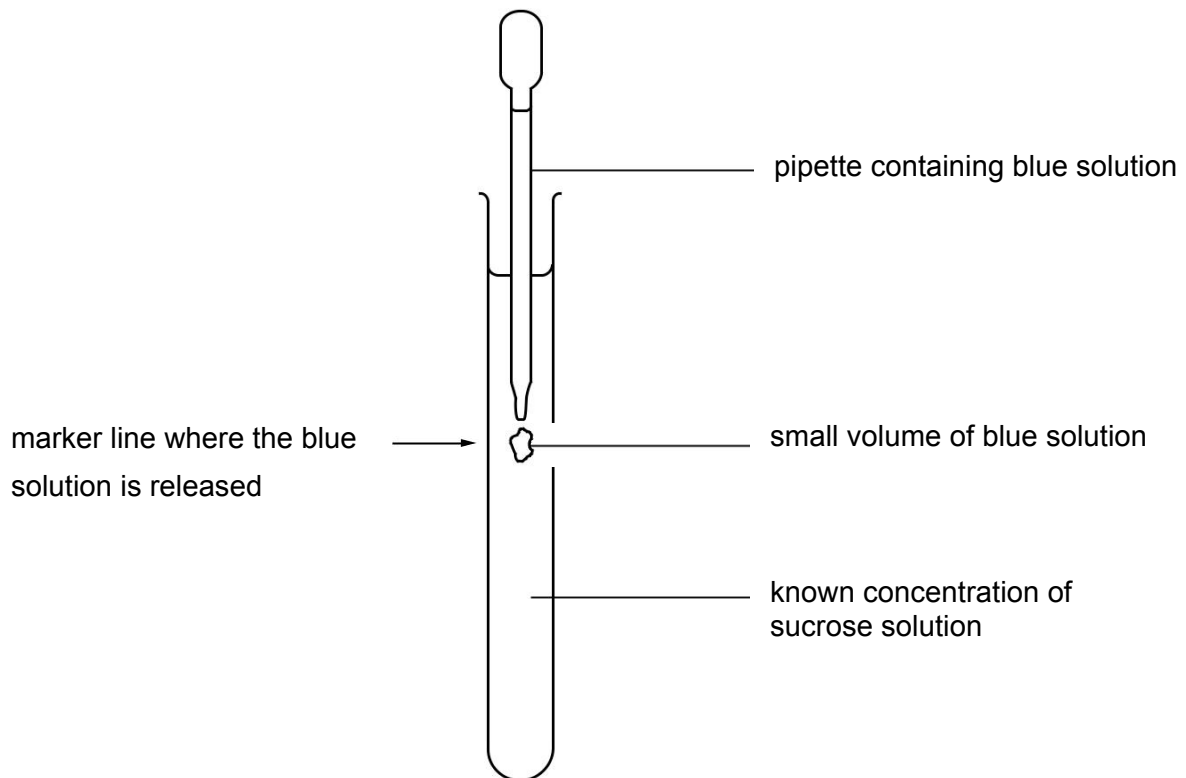


Fig. 1.2

The pipette is removed immediately after the blue solution is released.

The blue solution may move up, move down or remain at the same level depending on the difference in the concentration between the blue solution and the known concentration of sucrose solution.

(ii) Think about how the blue solution will move in the known concentration of sucrose in the test-tubes shown in Fig. 1.3.

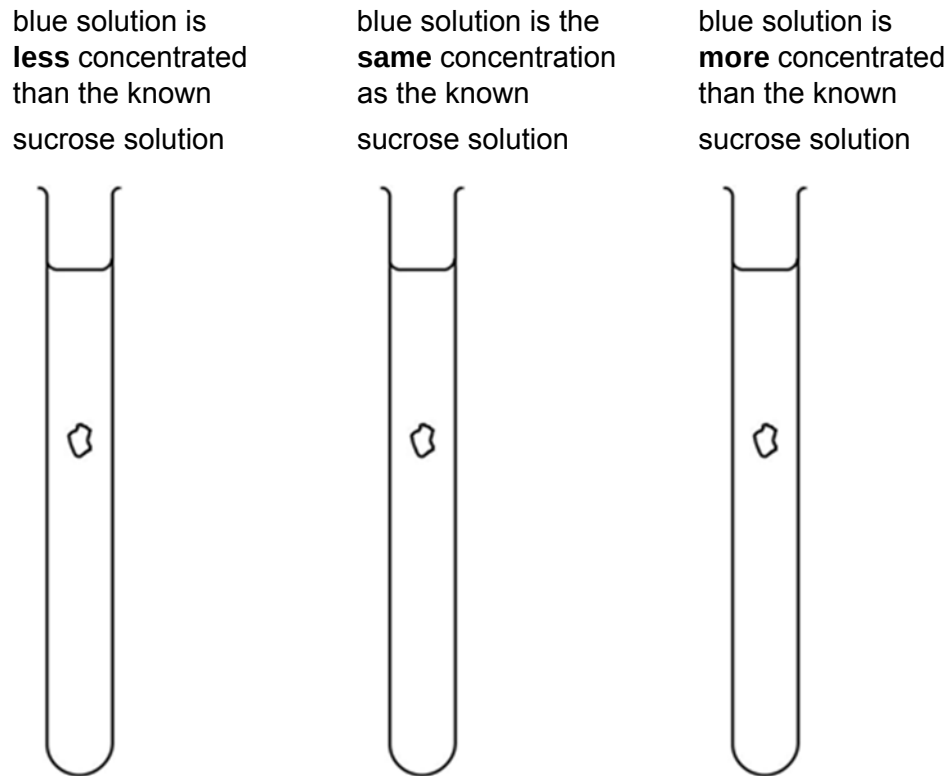


Fig. 1.3

Complete Fig. 1.3 by drawing an arrow on each test-tube, using the key, to show how you expect the blue solution to move.

key:	
↑	blue solution moves up
↓	blue solution moves down
→	blue solution remains at the same level

[1]

completes Fig. 1.3 drawing all three directions correctly (up + level + down) ;

(b) You are required to investigate osmosis in potato tissue so that you can estimate the water potential of the potato cells.

You are provided with:

labelled	contents	hazard	volume / cm ³
S	1.00 mol dm ⁻³ sucrose solution	none	200
W	distilled water	none	200

labelled	details
P	7 pieces of potato, cross sectional area 1.5 cm × 1.0 cm of variable length

You are required to make **simple** dilutions of the 1.00 mol dm⁻³ sucrose solution, **S**, which reduces the concentration between each dilution.

You will need to prepare 40 cm³ of each concentration.

Decide which concentrations of sucrose solution to prepare using simple dilutions of **S**.

(i) Complete Table 1.1 to show how you will prepare the other concentrations.

Table 1.1

final concentration of sucrose solution / mol dm ⁻³	volume of S / cm ³	volume of distilled water, W / cm ³
1.00	40	0
0.8	32	8
0.6	24	16
0.4	16	24
0.2	8	32

[3]

for at least 4 suitable concentrations of **S** ;

e.g. 0.8, 0.6, 0.4 and 0.2

decides correct volume of sucrose volumes for selected concentrations ;

decides correct total volumes (40 cm³) for each concentration ;

Read step 1 to step 11 before proceeding.

Proceed as follows:

1. Prepare all the concentrations of sucrose solution as shown in Table 1.1 in the beakers provided.
2. Measure 4.5 cm from the bottom of each **large** test-tube and put a mark, as shown in Fig. 1.4.

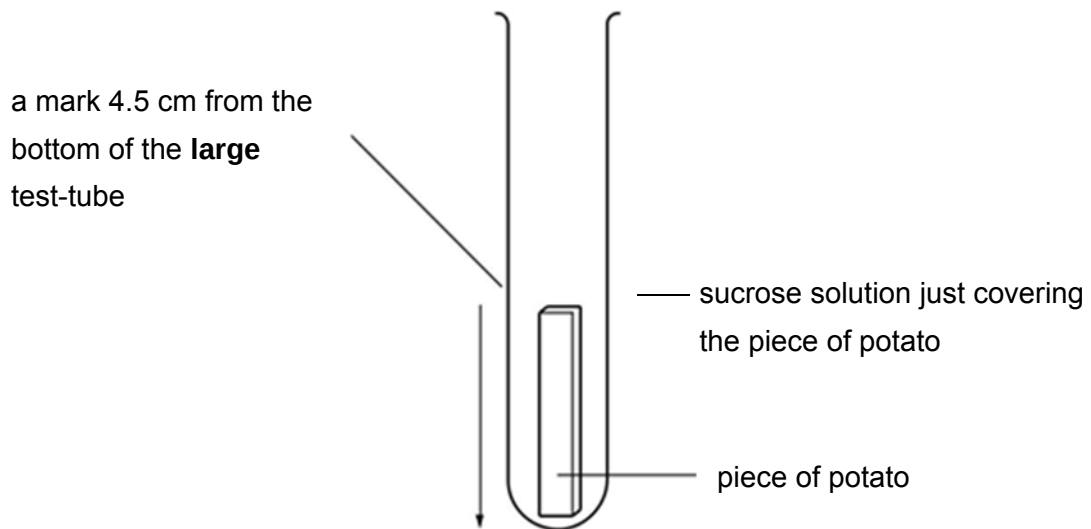


Fig. 1.4

- (ii) Decide on the length of each piece of potato you will use, so that the volume of sucrose solution just covers the piece of potato, as shown in Fig. 1.4.

State the length of the pieces of potato you will use.

length = [1]

decides appropriate length of potato pieces ;

e.g. 4.0 cm

3. Cut enough pieces of potato to put into the sucrose concentrations that you prepared in step 1.
4. Put the pieces of potato on a paper towel to remove any excess fluid.
5. Put one piece of potato into each of the large test-tubes from step 2.
6. Put 1.00 mol dm^{-3} sucrose solution, **S**, into a large test-tube up to the mark made in step 2.
7. Repeat step 6 with each of the other concentrations of sucrose solution.
8. Start timing and leave for 15 minutes.

While you are waiting for 15 minutes continue with step 9 to step 11 and continue with the other questions.

9. Measure 6 cm from the **top** of each of the **small** test-tubes and put a mark, as shown in Fig. 1.5.

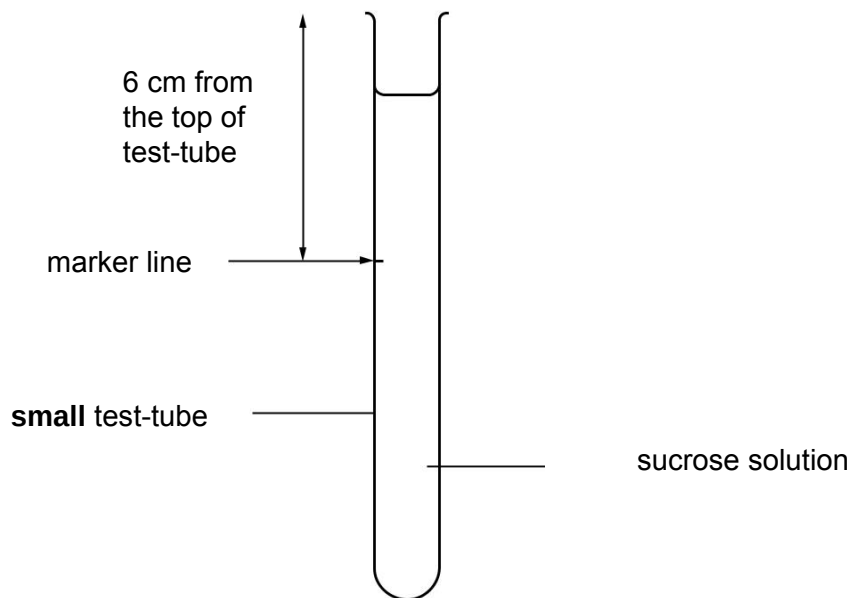


Fig. 1.5

10. Put 15 cm^3 of 1.00 mol dm^{-3} sucrose solution, **S**, into one of the **small** test-tubes from step 9.
11. Repeat step 10 with each of the other sucrose solutions that you prepared in step 1.

You are provided with:

labelled	contents	hazard	volume / cm^3
M	methylene blue solution	none	15

12. After leaving the pieces of potato for 15 minutes, put a drop of **M** into the **large** test-tube containing 1.00 mol dm^{-3} sucrose solution, **S**, and a piece of potato.
13. Gently shake the **large** test-tube to mix **M** with the sucrose solution.
14. Repeat step 12 and step 13 with each of the other concentrations of sucrose solution.

Read step 15 to step 19 before proceeding.

15. Use a pipette to remove a sample of the blue solution from the **large** test-tube containing 1.00 mol dm^{-3} sucrose solution, **S**.

*You will now use the **small** test-tubes as in Fig. 1.5.*

16. Put the end of the pipette into the **small** test-tube containing 1.00 mol dm^{-3} concentration of sucrose solution, **S**. This should be level with the marker line on the test-tube as shown in Fig. 1.2 on page 3.

17. Release a small volume of the blue solution, then immediately remove the pipette from the test-tube.

It is possible to repeat step 17 without having to replace this sucrose solution.

18. Immediately observe the direction **and** the speed of movement of the blue solution. *You are **not** required to measure the speed.*

Record these observations in **(b)(iii)**.

19. Repeat step 15 to step 18 using the other concentrations of sucrose solution. Make sure that the small volume of the blue solution from the **large** test-tube is put into the **small** test-tube labelled with the same concentration.

(iii) Record your observations of direction and speed of movement in an appropriate table.

- 1 table drawn + heading, concentration of sucrose solution / mol dm^{-3} ;
- 2 heading, direction of movement ;
- 3 records speed of movement in an appropriate way ;
- 4 decides to do repeated drops ;
- 5 results for at least 4 concentrations of sucrose ;
- 6 correct sequence of directions ;

- (iv) Using your results in (b)(iii) estimate the concentration of sucrose solution with a water potential equal to the water potential of the potato tissue.
correct estimate of concentration of sucrose according to results in (b)(iii) ;

.....
.....[1]

- (v) Identify **one** significant source of error in this investigation.
identifies one significant source of error ;
e.g. difficulty of measuring and cutting pieces of potato to correct dimensions

.....
.....
.....[1]

- (vi) Describe how you would use this procedure to produce a more accurate estimate of the concentration of sucrose solution with a water potential equal to the water potential of the potato tissue.

uses increased number of concentrations of sucrose solution ;
between 2 stated concentrations appropriate to candidate's results ;
read off from graph of results or replicate ;

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.....[3]

- (vii) Describe the movement of water when the concentration of the sucrose solution surrounding the piece of potato has a water potential equal to the water potential in the potato tissue.

no net movement of water or reference to dynamic equilibrium ;

.....
.....[1]

- (viii) Using the same procedure a student observed that the blue solution stayed in the same position (did not move up or down) in a concentration of sucrose solution of 0.3 mol dm^{-3} .

Table 1 shows the relationship between the concentration of sucrose solution and the water potential of the sucrose solution.

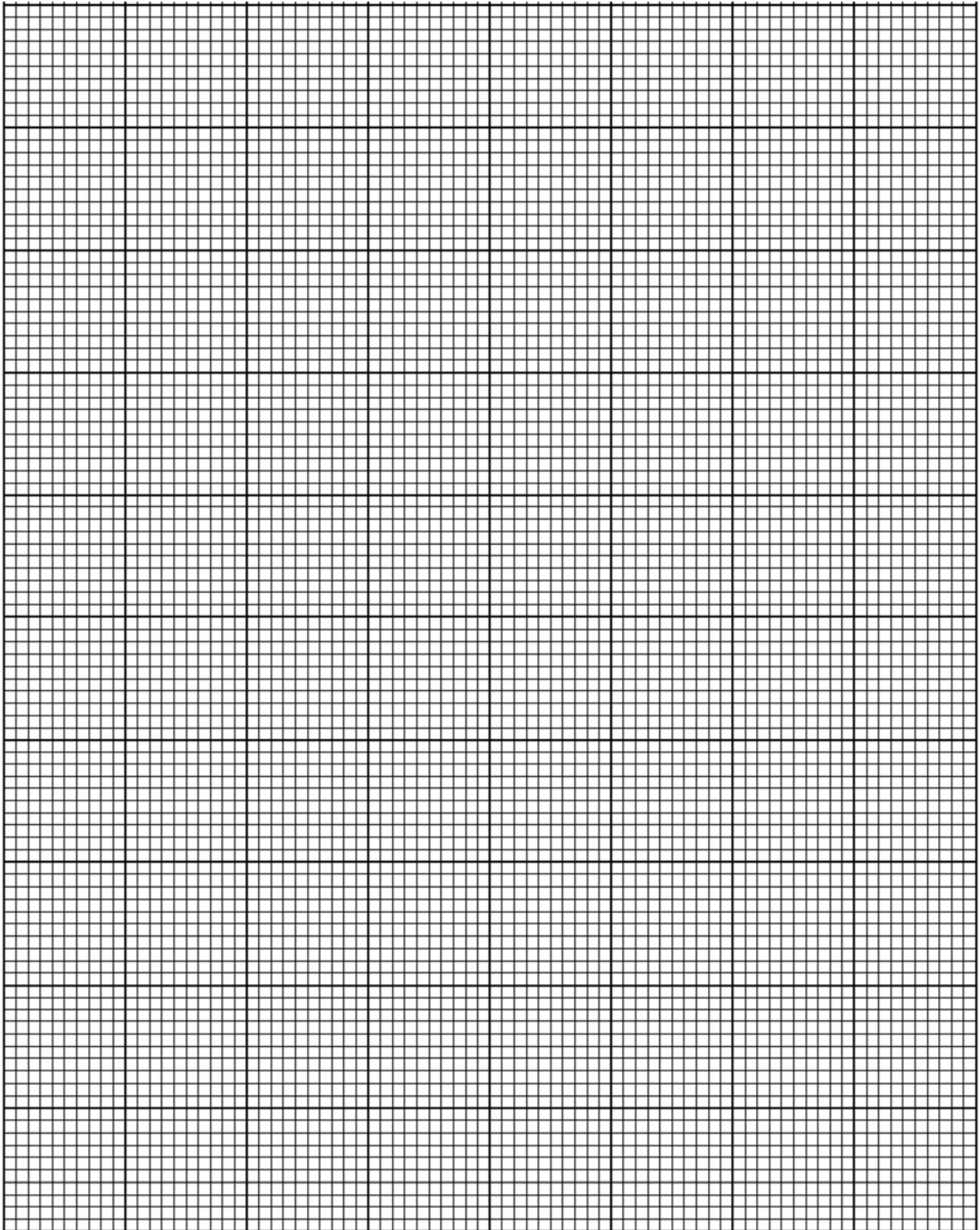
Table 1

Concentration of sucrose solution / mol dm^{-3}	Water potential / $\text{k Pa} \times 10^2$
0.2	-5.5
0.4	-11
0.6	-18
0.8	-27
1.0	-35

Plot a graph using the data in **Table 1**.

[4]

- 1 Correct orientation of the axes and correct labels with units;
x-axis: concentration of ascorbic acid solutions / mg cm^{-3}
y-axis: number of drops of ascorbic acid added
- 2 Appropriate scale and no awkward scale;
- 3 all points accurately plotted;
- 4 best fit **curve/line**;



[Total: 21]

- 2 You are provided with a quantity of vitamin C solution and a dye called DCPIP.

You are also provided with three test-tubes containing respectively lemon juice, orange juice, grapefruit juice, and labelled as such. These juices contain natural vitamin C and the dye DCPIP can be used to determine the concentration of this vitamin in the juices.

Proceed as follows:

- 1 Into a clean test-tube, transfer a quantity of the dye DCPIP to a depth of 0.5 cm. Take note its colour.
- 2 Fill a teat pipette with vitamin C solution. Add one drop of vitamin C solution to the DCPIP solution in the test-tube and shake gently. Continue to add the drops, counting the number of drops which are needed to bring about a colour change. Shake gently after each drop, refilling the pipette if necessary.
- 3 Record the initial colour of DCPIP (from step 1) and the first colour change after vitamin C is added as well as the number of drops counted to bring about this colour change in a suitable table.
- 4 After the first colour change, continue adding drops of vitamin C and counting the drops until the DCPIP solution becomes colourless/or consistent pale yellow. (Ignore any coloured granules that might form.). Record the number of drops counted in the same table from step 3.
- 5 Repeat steps 1 to 4 adequately to obtain enough data for analysis, cleaning all apparatus before use.
- 6 Place the DCPIP solution into each of three clean test-tubes to a depth of 0.5 cm. (The amount of DCPIP solution must be exactly the same in each of the tubes). Label the tubes A, B and C.
- 7 Fill a clean teat pipette with lemon juice and drop by drop add this to the contents of tube A, shaking the tube gently after each drop. Count the number of drops needed to turn the DCPIP solution colourless. Repeat this step adequately to obtain enough data for analysis.
- 8 Repeat the step 7 with orange juice and grapefruit juice, using a clean pipette each time to add the juice to the DCPIP solution in tubes B and C respectively.

9 Record the results for the three juices in an appropriate table. [4]

Solution	Replicates	Initial colour of DCPIP	First colour change	Drops required to achieve 1 st colour change	Second Colour Change	Drops required to achieve 2 nd colour change
Vitamin C	1	Blue	Brown/Yellow	1	Pale Yellow	2-5
	2					
	Average					
Lemon Juice						
Grapefruit juice						
Orange juice						

Table with headings – 1

Results for Vitamin C and juices with at least one replicate – 1

Trend for Juices with at least one replicate – 1 (Lowest drops (Orange) to highest drops (Lemon))

Calculation of average for each solution - 1

10 What conclusions can you draw from your results?

- Relate trend to relative concentrations of Vitamin C
- Higher concentrations of Vitamin C = lower pH = more H⁺
- DCPIP reduced faster = faster colour change

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.....[3]

- 11 Comment on the main source(s) of error and the limitations of the measurements or experimental procedure.

identifies three of:

- uneven mixing of the juice with DCPIP solution;
- measuring DCPIP to a depth of 0.5 cm in test-tube;
- inconsistent drop size and release of drop from teat pipette;
- use of ruler to mark out a depth of 0.5 cm on test-tube;
- visual comparison of colour change (when DCPIP decolourises);
- varying drop size of vitamin C (or juice) due to way and the speed at which it is released from teat pipette;
- viscosity of fruit juice affecting size of the drop being formed before it is released from the teat pipette;

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.....[3]

- 12 What improvements could you make to the experimental procedures to overcome these sources of error?

Shake before using juice

Use colourimeter to determine colour change

use a syringe / graduated pipette to transfer a known volume of DCPIP;

use a burette to dispense equal sized drops / regulate the release the fruit juice;

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.....[2]

- 13 In this practical, you are required to investigate the effect of different wavelength of light on a leaf extract using a solution of the dye DCPIP (dichlorophenol indophenol). DCPIP turns from blue to colourless when it is reduced. The leaf extract will be prepared by you.

You may select from the following apparatus and use appropriate additional apparatus:

- Petri dish
- 6 melting point capillary tubes – 10 cm long checked to make sure both ends are open
- Plastic specimen tube (minimum 3 cm x 1 cm)
- Glass rod that fits the specimen tube
- White tile
- Sharp knife or scalpel
- One sheet of Aluminium foil, approximately 5 cm x 5 cm for capillary tube
- One sheet of Aluminium foil, approximately 12 cm x 12 cm to cover the Petri dish completely
- Bench lamp with 60W bulb
- Stopwatch
- 5-cm³ syringe
- Dropper pipette
- Paper towels
- Distilled water in a wash bottle
- 20 cm³ buffer solution dispensed in a Styrofoam cup containing crushed ice
- 10 cm³ DCPIP solution dispensed in a Styrofoam cup containing crushed ice
- 40 cm² fresh leaf (spinach), roughly 10 cm x 4 cm
- 2 cm x 5 cm green and red transparency paper

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of result tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[9]

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Mark scheme for your reference

Theoretical background [2m max]

E: Electrons are emitted in the presence of light / during photophosphorylation from the leaf extract

BC: Reduce the DCPIP which is in the mixture, turning it from blue to colourless.

WL: Different wavelength of light will result in different rate of photosynthesis;

ES: Reducing the enzyme-substrate complexes formed per unit time;

Dependent variable: time taken for decolouration

Independent variable: wavelength of light (red and green)

Constant variables: temperature, volume of leaf extract / DCPIP

Hypothesis: Red light will result in a higher rate of photosynthesis compared to green light.

Procedures [8m max] – A SCAR

CV1: Fixed volume / concentration of cold buffer solution

CV2: Fixed distance from lamp from leaf extract;

CV3: Fixed volume of DCPIP / leaf extract

AD: Diagram of experimental set-up (with lamp, white tile and tubes of leaf extract);

G: Grind the chopped leaf to obtain a green leaf extract;

D: Decant the leaf extract;

L: Label five capillary tubes;

AL: Cover the Petri dish with aluminium foil to shield the contents from light

S: Colour standard (leaf extract without DCPIP);

C: Control by covering tube of leaf extract with aluminum foil

R: Repeat experiment three times + calculate mean time;

TB: tabulation of data with correct headings and units;

Recommended Procedures

1. Place the fresh leaf on a white tile. With a scalpel, carefully remove any large veins and discard them.
2. Chop the rest of the leaf into small pieces and place the chopped leaf into a plastic specimen tube.
3. Using a syringe, add 2 cm³ of cold buffer solution into the specimen tube.
4. Carefully grind the chopped leaf in the specimen tube using a glass rod for about 1 minute to obtain a green leaf extract.
5. Clean the surface of the white tile with a paper towel.
6. Place the Petri dish on the edge of the tile, so that the dish is tilted.
7. Decant the leaf extract from the specimen tube into the Petri dish so it forms a small puddle.
8. Completely cover the Petri dish with aluminium foil to shield the contents from light. **Leave the foil in position except at times when removing samples.**
9. Fold both the red and green transparency paper in half along their length to produce a roof structure
10. Label five capillary tubes from 1 to 5.
11. Take a capillary tube and stand one end in the leaf extract. Some liquid will rise up into the capillary tube (5 cm).
12. Place this capillary tube containing the green leaf extract on the white tile labelled as **tube 1**, which will act as a colour standard.
13. Using a dropper pipette, add **5 drops** of 1% solution DCPIP to the leaf extract in the Petri dish.
14. Rock the Petri dish to mix the liquids and **replace the foil cover**.
15. Take a new capillary tube. Stand one end of each tube in the leaf extract/DCPIP mixture to collect some of the liquid and cover capillary tube quickly in aluminium foil to prevent exposure to light. Do likewise for the other capillary tubes.
16. Place these tubes labelled as **tube 2 - 5** on the white tile, as follows:
 - 2: under red filter paper
 - 3: under green filter paper
 - 4: under white light
 - 5: under aluminium foil
17. Position the lamp 10 cm away from the capillary tubes. (Refer to Fig. 1)

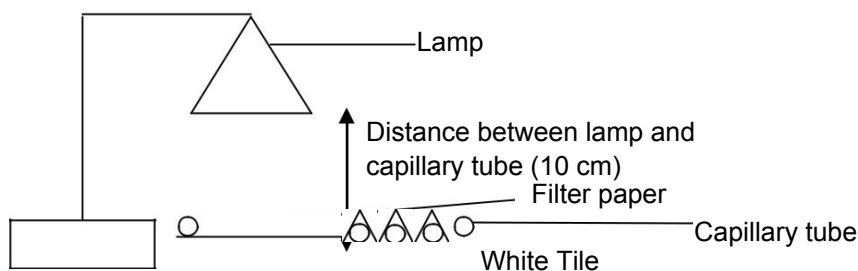


Fig. 1

18. Quickly remove the aluminium foil from capillary tubes, turn on the lamp and start timing using a stopwatch. Record the timing when DCPIP turns colourless (it should appear the same colour as the colour standard).
19. Repeat steps 15 to 17 to twice.

20. Record all readings in the table below and calculate the mean time taken.

Data Presentation

Table of Results of Time Taken for DCPIP to decolourise

Tube	Time taken for DCPIP to decolourise / s			
	1 st attempt	2 nd attempt	3 rd attempt	Mean
2 (Red filter)				
3 (Green filter)				
4 (White light)				

21. Conclusion: The rate of photosynthesis is affected by wavelength of light. Certain wavelength of light gives a higher rate of photosynthesis. The rate of photosynthesis is highest under white light followed by red light, green light and no light. The wavelength of light in the red spectrum will lead to a faster rate of decolourisation DCPIP due to higher rate of photosynthesis which produces electrons at a higher rate. White light contains both red and blue light which gives a higher rate of photosynthesis. Tube 5 functions as a control.

Risk assessment [2m max]

RA15: DCPIP is an irritant and gloves should be worn to avoid contact with skin and eye.

RA16: Keep the scalpel in the basket when not using it, so as to prevent accidental cuts & injuries.

RA17: Wipe off any spillage of solutions, so as to prevent injuries due to accidental slippage.

Examiner's Comments: (if any)

For risk assessment, only specific risks pertaining to experiment is accepted.

Many candidates wrote 'sec' instead of 's' in their table of results.

Prevention of contamination of the equipment i.e. the white tile by wiping with paper towel

- 3 (a) **M1** is a slide of a stained transverse section through a plant root.

You are not expected to be familiar with this specimen.

Use a sharp pencil for drawing.

- (i) Draw a large plan diagram of a quarter of the root on **M1**, shown by the shaded area in Fig. 3.1. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams.

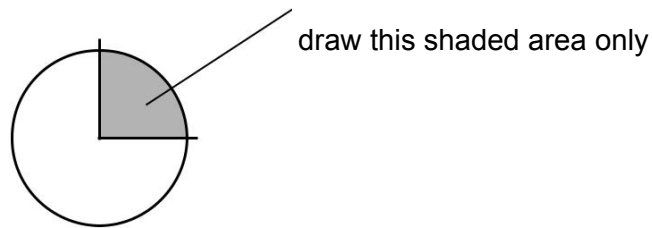


Fig. 3.1

Your drawing should show the correct shape and proportions of the different tissues.

- 1 plan diagram of appropriate size + no cells + no shading ;
- 2 correct section drawn + draws at least 3 different layers of tissue ;
- 3 draws 3 layers of tissue for the central stele or for the edge of the root ;
- 4 draws air spaces in the cortex if there are;
- 5 uses one label line + one label to identify the endodermis ; **(do we want to test students this?)**

(1M each)

(ii) Observe the central tissue in the root on **M1**.

Select **one** large xylem vessel and **three** adjacent (touching) smaller xylem vessels from the tissue at the centre of the root. The large xylem vessel must touch each of the three smaller xylem vessels.

Make a large drawing of this group of **four** xylem vessels.

- 1 quality of line for the outer wall of xylem vessels + cells of appropriate size ;
- 2 draws only four xylem vessels + with the large xylem vessel touching each of the other 3 smaller vessels ;
- 3 cell walls drawn as two lines close together ;
- 4 draws the largest xylem vessel lumen at least twice the size of the smallest xylem vessel lumen ;

(1m each)

[4]

- (b) Fig. 3.2 is a photomicrograph of a stained transverse section of a different root. You are not expected to be familiar with this specimen.

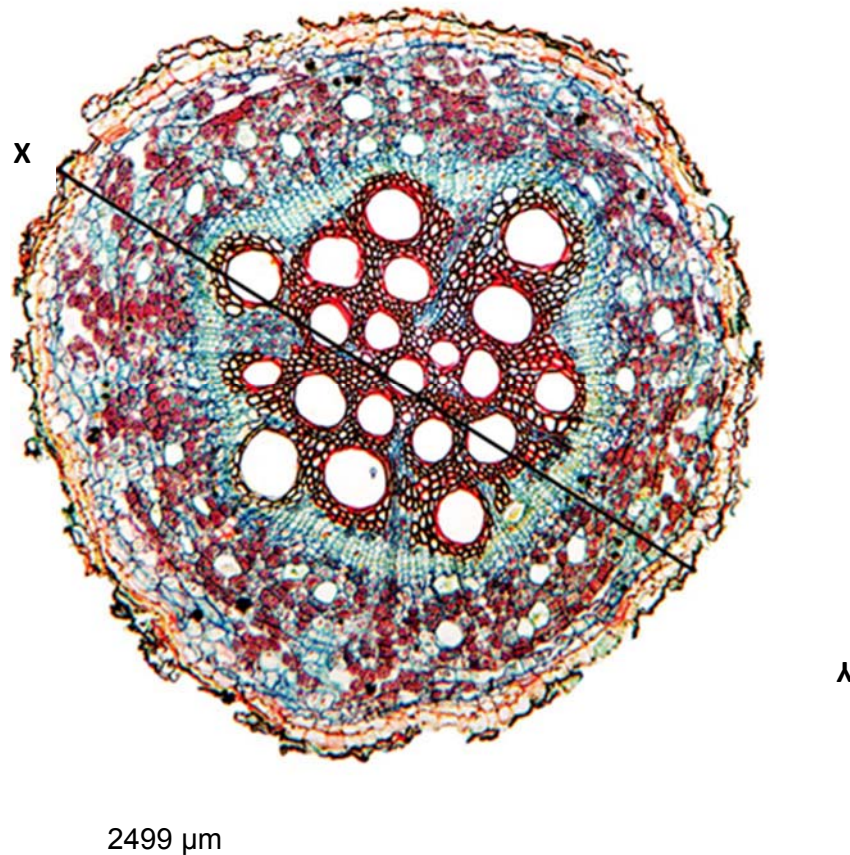


Fig. 3.2

In Fig. 3.2 the line **X–Y** is drawn across the diameter of the root.

Use the line **X–Y** and the scale bar to calculate the actual diameter of the root.

You may lose marks if you do not show your working or if you do not use appropriate units.

- 1 correct measurement of scale bar ;
- 2 shows length of scale bar in μm , divided by 2499 ;
- 3 shows length of line, **X–Y**, divided by answer to mp2 ;
- 4 decides to record answer in μm ;

alternative ways to calculate actual diameter accepted

actual diameter = [3]

- (c) Observe the root on **M1** and the root in Fig. 3.2 and identify the differences between them. Record the observable differences in Table 3.1.

Table 3.1

feature	M1	Fig. 3.2

[2]

[Total: 13]